

Jairaj Narendran

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Education

Boston University (BU)

Boston, MA

Ph.D., Materials Science and Engineering

Expected May 2026

Mechanics of Slender Structures (MOSS) Lab

Advisor: Dr. Douglas P. Holmes

M.S., Materials Science and Engineering

January 2026

University of Illinois Urbana-Champaign (UIUC)

Champaign, IL

B.S., Bioengineering, Minor in Materials Science and Engineering

May 2021

Research Experience

Boston University

Boston, MA

Graduate Researcher – [DARPA Bio-inspired Restoration of Aged Concrete Edifices \(BRACE\)](#)

September 2024 – Present

- Design, develop, and verify fiber-confined construction material processes for airfield damage repair – from initial material selection and fiber sleeve fabrication through pilot-scale assembly testing – with the goal of creating a reliable, field-deployable system.
- Apply statistical tools to evaluate building materials for compressive performance and durability in compliance with ASTM C39 standards, interpreting test results to recommend material selections and process parameters.
- Develop design-for-manufacturing workflows for knit fiber sleeve fabrication across multiple high-performance fiber types, optimizing construction processes for repeatability and rapid field deployment.
- Drive iterative process refinement through hands-on experimentation and root cause analysis, ultimately exceeding the 400 PSI structural performance milestone by over 200%.
- Coordinate across research institutions and government stakeholders to keep the program on track, delivering monthly written reports and biweekly presentations summarizing progress, identifying risks, and outlining next steps.
- Provide technical oversight during construction and assembly activities, setting expectations and ensuring processes are executed to specification.

Boston University

Boston, MA

Graduate Researcher – MOSS Lab

September 2021 – Present

- Translated user needs into design specifications for a robotic kirigami gripper for people with impaired dexterity, developing and verifying successive prototype generations through a \$75,000 BU Ignition Award – hitting performance milestones including an 8X improvement in fatigue life and a 6X increase in load capacity on time and within budget.
- Led prototyping and testing efforts using parametric CAD (OnShape) and 3D printing (FDM, SLS), applying DFM principles from early concept stages through validated functional prototypes.
- Built Python scripting and MATLAB pipelines to process experimental datasets, extract performance metrics, and support data-driven decision making across iterative development cycles.
- Designed and executed mechanical test programs on an Instron dynamic testing machine, applying statistical analysis to results and writing reports connecting design variables to measurable performance outcomes.
- Characterized elastogranular bilayer shell structures exhibiting a tunable, reversible 3.5X increase in stiffness and 2X increase in critical buckling load, developing quantitative process-property models linking fabrication parameters to system-level performance.
- Awarded Best Presentation Prize at NEW.Mech 2023 for research on snapping kirigami shell mechanics.

University of Illinois Urbana-Champaign

Champaign, IL

Undergraduate Researcher

September 2018 – May 2021

- JI Labs (2020-2021): Characterized structural and adhesion properties of engineered nanofiber scaffolds using super-resolution nanoscopy, producing structured datasets for materials evaluation and documentation in a biomedical research context.
- PhantomCor (2019-2021): Developed and iteratively refined material processing workflows to achieve physiologically-relevant mechanical and physical properties, applying systematic process development methodology to meet defined performance specifications for surgical simulation devices.

- MTM Lab (2019): Developed an automated image processing pipeline using ImageJ and CellProfiler for high-throughput biological image analysis, achieving a 10X improvement in processing rate – experience transferable to vision-based inspection and quality monitoring in manufacturing contexts.
- MatMed Lab (2018-2019): Synthesized and characterized carbon nanomaterial systems and graphene-based composite coatings, implementing new materials processing techniques to develop functional systems for diagnostic applications.

Leadership

Boston University

Boston, MA

Research Mentor and Graduate Student Teacher

September 2021 – Present

- Instructed 80+ mechanical engineering students in instrumentation, data acquisition, signal conditioning, circuit design, and controls fundamentals using LabVIEW and MATLAB, receiving 4.6 out of 5.0 on student evaluations.
- Mentored 10+ master's and undergraduate students in experimental design, data analysis, and scientific communication to develop independent research skills.
- Developed curriculum for and mentored 10+ high schoolers through Upward Bound to give STEM exposure to students from low-income families and groups traditionally underrepresented in postsecondary education.

Boston University

Boston, MA

Peer Mentorship Program Director – MSE-GSA BU Chapter

April 2023 – May 2025

- Organized and ran programming for 50+ graduate students, coordinating across faculty, department staff, and industry partners while handling logistics, scheduling, and budgeting.
- Hosted networking events connecting 35+ students with representatives from materials science, engineering, and manufacturing companies in the Boston area.

University of Illinois Urbana-Champaign

Champaign, IL

Technical Director – Biomedical Engineering Society UIUC Chapter

April 2018 – May 2021

- Mentored 15+ undergraduate engineering project team leads on effective leadership techniques, gathering resources, and managing project budgets through 1-on-1 monthly meetings.
- Managed ~\$4,000 annual budget across 20 engineering project teams and implemented a shared inventory system to increase resource efficiency.
- Organized technical workshops on valuable engineering tools such as Python, MATLAB, and ImageJ with consistent turnouts of 10+ students.

Skills

- **Mechanical Design & Prototyping:** 3D Printing (SLS, FDM), belt sander, CNC mill, GD&T, laser cutter, lathe, miter saw, silicone molding
- **Technical:** Adobe Illustrator, Arduino IDE, Blender, CAD (PTC OnShape), ImageJ, Java, LabVIEW, MATLAB, Office 365, Python
- **Testing & Verification:** DAQ machine, DIC, digital multimeter, dynamic testing machine (Instron), lightweight deflectometer, oscilloscope, signal generator

Publications and Presentations

- **J. Narendran**, A.A. M. Campbell, K. Wood, J. Roman, D.P. Holmes (2026). Knit Sleeves for Load Bearing Elastogranular Structures. *In Preparation*
- **J. Narendran**, P. Limcaoco, S. Basti, D.P. Holmes (2026). Compliance-Shifting Elastogranular Shells. *In Preparation*
- A.A. M. Campbell, **J. Narendran**, K. Wood, J. Roman, D.P. Holmes (2026). Brick-like Elastogranular Metamaterials: Adhesive-free Load-bearing Columns. *In Preparation*
- **J. Narendran**, P. Limcaoco, S. Basti, D.P. Holmes. Compliance-Shifting Structures using Elastogranular Shells. *APS Global Physics Summit 2025*, Anaheim, California. March 16-21, 2025
- **J. Narendran**, D.P. Holmes. Rapid Closure of Kirigami Grippers. *NEW.Mech 2023*, Boston, Massachusetts. May 26, 2023
- W. Liu, A. Padhi, X. Zhang, **J. Narendran**, M.A. Anastasio, A.S. Nain, J. Irudayaraj (2022). Dynamic Heterochromatin States in Anisotropic Nuclei of Cells on Aligned Nanofibers. *ACS Nano*, Vol. 16, Issue 7, pp. 10754-10767. DOI: 10.1021/acsnano.2c02660

- L. Garcia, G. Kerns, K. O'reilley, O. Okesanjo, J. Lozano, **J. Narendran**, C. Broeking, X. Ma, H. Thompson, P.N. Njeuha, D. Sikligar, R. Brockstein, H.M. Golecki (2022). The Role of Soft Robotic Micromachines in the Future of Medical Devices and Personalized Medicine. *Micromachines*, Vol. 13, Issue 28, pp. 1-25. DOI: 10.3390/mi13010028